CHATFIELD RESEARCH

QUANTITATIVE RESEARCH DIVISION

Project Week 1 Review and Week 2 Plan

Forecasting UK Labour-Market Vacancies using Economic and Job-Posting Data

Research Progress • Week 2 Assignments • Project Roadmap • Final Outputs

Prepared for internal circulation

27 October 2026

Project Overview and Objectives

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1 Research Question

Can job-posting data improve forecasts of UK labour-market vacancies?

- We investigate whether Indeed job-posting data can provide useful, timely information for forecasting UK vacancies.
- We compare a univariate ARIMA model with an ARIMAX model that includes Indeed data.
- Our focus is on out-of-sample forecasting performance and the economic interpretation of the results.

2 Our Objectives

1. **Build a high-quality dataset**
   Collect and prepare ONS vacancy data and Indeed job-posting data.
2. **Develop and compare forecasting models**
   Estimate an ARIMA model and an ARIMAX model using real UK data.
3. **Evaluate forecasting performance**
   Compare models using out-of-sample accuracy metrics and diagnostic tests.
4. **Produce a joint research output**
   Publish a written research paper and a reproducible GitHub repository.

3 Key Details

Topic	Forecasting UK labour-market vacancies using economic and job-posting data
Timeframe	Six-week research project (Weeks 1–6)
Data sources	ONS vacancies (seasonally adjusted) Indeed job-posting index
Models	ARIMA (univariate) ARIMAX (with Indeed data)
Outputs	Joint research paper Public GitHub repository
Contributors	Head of Quant + 3 analysts (four equal authors)

Project Structure and Workstreams

Three specialised workstreams, one research project and four equal authors.

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1 Quant 1 — Economic Data

Collect, prepare and document ONS and Indeed data.

Build a clean, aligned dataset of UK vacancies and job-posting activity to be used for modelling and analysis.

Key outputs

- *ons_monthly.csv* (ONS vacancy series)
- ons_indeed_aligned.csv (matched dataset)
- Data preparation script
- Data dictionary and notes

Contribution to the project

Provides the economic data and supporting analysis for the research paper and repository.

2 Quant 2 — Statistical Methodology

Identify and justify an ARIMA specification.

Apply statistical tests, compare candidate models and check diagnostics using real ONS data.

Key outputs

- Python script with statistical tests
- Model comparison table (AIC/BIC)
- Stationarity and diagnostic plots
- Methodological report with recommended ARIMA specification

Contribution to the project

Provides the statistical justification for our forecasting methodology and model selection.

3 Quant 3 — Forecasting and Evaluation

Generate and evaluate real-data forecasts.

Implement ARIMA forecasts, compare with a naive benchmark and produce out-of-sample performance metrics.

Key outputs

- Updated Python forecasting script
- forecasts.csv (actual and predicted values)
- accuracy.csv (MAE and RMSE results)
- Forecast comparison figure and README

Contribution to the project

Produces the empirical forecasting results and evaluation framework for the research paper.

Joint output

All four contributors will be credited as authors of a single research paper and a public GitHub repository.

Research paper
Scientific-style working paper

GitHub repository
Fully reproducible research code

Six-Week Research Plan

From data collection to a jointly authored research paper and GitHub repository.

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Week 1
Research Foundations

Establish the research foundations.

- Explore ONS vacancy data
- Initial data analysis and visualisation
- Review forecasting methodology literature
- Build a prototype ARIMA forecasting model (using simulated data)

Completed

Week 2
Data Preparation and Initial Modelling

Prepare real data and initial modelling.

- Finalise ONS and Indeed datasets
- Investigate stationarity and select an ARIMA specification
- Generate first real-data forecasts and compare with a naive benchmark

Current focus

Week 3
ARIMAX Model Development

Incorporate Indeed data and develop ARIMAX.

- Integrate Indeed job-posting index
- Build and estimate ARIMAX model
- Compare ARIMA and ARIMAX specifications
- Prepare updated forecasting pipeline

Next

Week 4
Results & Statistical Evaluation

Evaluate model performance.

- Compare forecast accuracy (MAE, RMSE)
- Conduct diagnostic tests and robustness checks
- Assess economic significance of results
- Analyse key findings

Week 4

Week 5
Research Paper Development

Write and integrate the research paper.

- Combine all workstreams into a full paper
- Write introduction, methodology, results and discussion
- Create figures and tables
- Finalise references and structure

Week 5

Week 6
Final Review and Publication

Finalise and publish our outputs.

- Complete final review and edits
- Ensure reproducibility of all code and results
- Prepare public GitHub repository
- Finalize and publish the research paper

Week 6

End goal
A high-quality, reproducible research project with a scientific paper and public GitHub repository.

Research paper
Jointly authored working paper

GitHub repository
Fully documented and reproducible code

Week 1: Overall Research Progress

Three independent research foundations established, ready for real-data analysis.

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1 Economic Data

ONS vacancies and Indeed job-postings

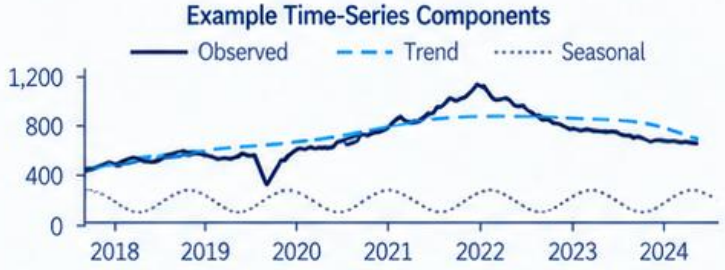
- Collected the ONS UK vacancy series (seasonally adjusted).
- Sourced the Indeed job-posting index for the UK.
- Explored trends, volatility and key economic events (e.g. COVID-19).
- Developed initial alignment between the daily Indeed series and monthly ONS observations.



2 Statistical Methodology

ARIMA and ARIMAX framework

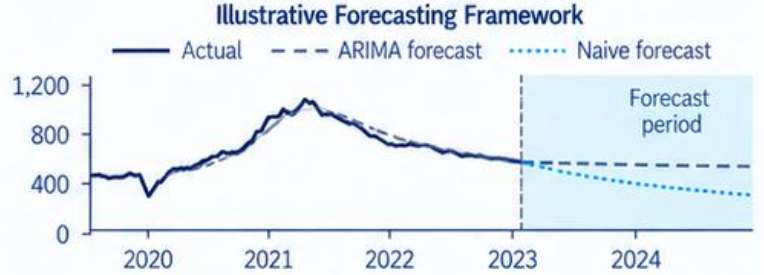
- Reviewed ARIMA and ARIMAX modelling approaches.
- Examined stationarity testing (ADF and KPSS).
- Discussed model selection using AICc and BIC.
- Considered model diagnostics (e.g. Ljung–Box, residual analysis).
- Outlined appropriate forecasting evaluation methods (e.g. rolling origin, MAE, RMSE, MASE).



3 Python Forecasting

Initial implementation and evaluation

- Developed a working ARIMA prototype using synthetic data.
- Implemented a naive forecasting benchmark.
- Calculated forecasting error metrics (e.g. MAE and RMSE).
- Created modular Python code with clear structure and documentation.
- Produced initial visualisations of forecasts and model performance.



Key Takeaway

In Week 1 we have established the data, statistical foundation and computational framework required to begin real-data forecasting. We are now ready to apply these components to the ONS vacancy series and move towards a full empirical analysis.

- Research foundations established
- All three workstreams making strong progress
- Ready to proceed to real-data modelling in Week 2

Quant 1 — Week 1 Review

Economic Data Collection and Exploration

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1 Key Tasks Completed

Established the economic data foundation.

Collected key datasets

ONS UK vacancy series (seasonally adjusted) and Indeed job-posting index for the UK.

Initial exploratory analysis

Investigated trends, volatility and key economic events (e.g. COVID-19); produced time-series visualisations.

Data alignment

Developed initial alignment between the daily Indeed series and the monthly ONS observations.

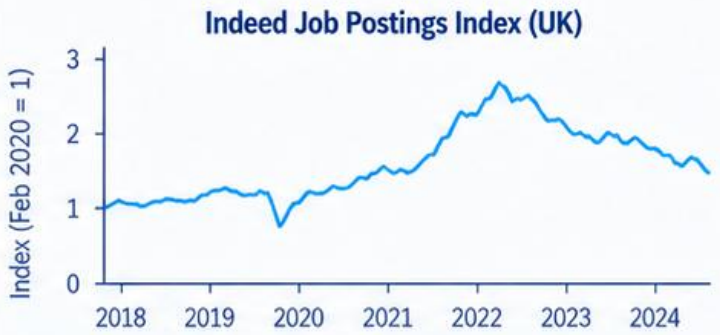
Preliminary investigation

Explored the relationship between changes in the two series and considered potential predictive value.

2 Example Exploratory Analysis

UK vacancies and Indeed job postings.





3 Strengths and Development Points

Summary of Week 1 performance.

Key strengths

- Successfully collected and documented both ONS and Indeed datasets.
- Produced clear exploratory analysis and visualisations.
- Demonstrated understanding of the different frequencies and reference periods.
- Made good progress on aligning the two series.
- Identified interesting patterns and economic context (e.g. COVID-19).

Development points

- Finalise and standardise the aligned dataset for use by the whole team.
- Ensure consistent handling of the ONS three-month reference periods and end-month labelling.
- Provide clear documentation of data sources, transformations and limitations.
- Focus on reproducibility and a clean, well-structured dataset for Week 2.

Overall assessment

Quant 1 has established the economic data foundation required for the projiect. The data collection and exploratory analysis provide a strong starting point for real-data modelling in Week 2.

Quant 1 — Week 2 Assignment

Finalise the research datasets for modelling and analysis.

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1 Objectives for Week 2

Produce clean, well-documented datasets.

Finalise the datasets

Clean and align the ONS vacancies and Indeed job-posting data for use by the whole team.

Ensure reproducibility

Provide a clear Python script to generate the datasets from the original sources.

Document methodology

Explain all data sources, transformations, frequency conversions and limitations.

Support team modelling

Produce datasets in a format that can be used directly by Quants 2 and 3 for modelling and forecasting.

2 Specific Tasks

Build on the Week 1 work.

Task	Details
Clean ONS vacancy data	• Use the seasonally adjusted series. • Check for any missing values or revisions. • Ensure consistent end-month labelling (three-month reference periods).
Prepare Indeed job-posting data	• Aggregate the daily Indeed index to monthly frequency (e.g. monthly average). • Check for outliers and structural breaks. • Document any adjustments made.
Align the datasets	• Match the time periods for ONS and Indeed data. • Ensure consistent date formatting and frequency. • Create a final merged dataset.
Documentation and reproducibility	• Write a data dictionary explaining sources, transformations and definitions. • Provide a fully reproducible Python script with clear comments. • Include notes on limitations and data availability timing.

3 Expected Outputs

To be submitted by Monday.

Output	Description	Format
ONS vacancy dataset	Clean, seasonally adjusted UK vacancies series	ons_monthly.csv
Indeed job-posting dataset	Monthly aggregated Indeed index	indeed_monthly.csv
Aligned dataset	Combined dataset with matching time periods	ons_indeed_aligned.csv
Data preparation script	Python script to reproduce all data processing steps	data_preparation.py
Data dictionary	Description of sources, transformations and key variables	data_dictionary.pdf

4 Key Considerations

Points to keep in mind.

- Use the full available history for both series (ONS and Indeed).
- Ensure correct handling of the ONS three-month reference periods.
- Check for any data revisions or methodological changes.
- Be explicit about how the Indeed index is aggregated to monthly data.
- Document any limitations or known issues.
- Ensure the final datasets are easy for the team to use in modelling and forecasting.

Deadline

Submit all outputs by **Monday 2 November 2026, 18:00**

Success criteria

Clean, aligned, well-documented datasets ready for Quants 2 and 3 to use in their Week 2 modelling and forecasting work.

Quant 2 — Week 1 Review

Statistical Modelling, Methodology and Model Selection Framework

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1 Key Tasks Completed

Established the statistical framework for the project.

Reviewed ARIMA and ARIMAX methods

Analysed the theoretical framework for univariate ARIMA and ARIMAX models, including assumptions and applications to economic time series.

Stationarity analysis

Examined ADF and KPSS tests, differencing requirements and the implications for model specification.

Model selection and diagnostics

Discussed AICc and BIC for model comparison and reviewed residual diagnostics (e.g. Ljung–Box test, residual ACF).

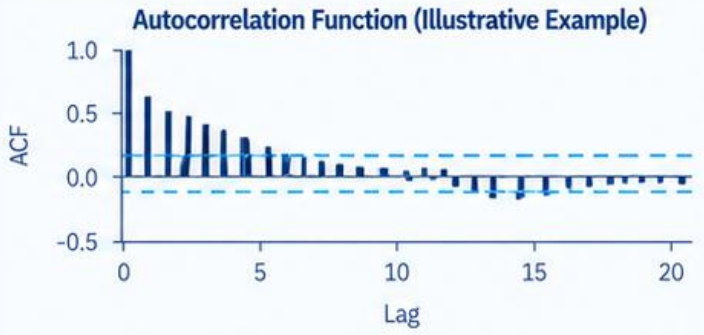
Forecasting evaluation framework

Outlined rolling-origin evaluation, naive benchmarks and key performance metrics (MAE, RMSE, MAPE, MASE).

2 Example Analysis (Illustrative)

Applying the methodology to a typical time series.





3 Strengths and Development Points

Assessment of Week 1 performance.

Key strengths

- Provided a clear and well-structured overview of the statistical methodology.
- Demonstrated good understanding of stationarity testing and model diagnostics.
- Considered appropriate model selection criteria (AICc and BIC).
- Addressed out-of-sample evaluation methods and elevant performance metrics.
- Produced clear notes and explanations that can be applied to the real data.

Development points

- Apply the methodology to the actual ONS vacancy series.
- Move from theoretical discussion to empirical model selection.
- Keep model selection separate from the holdout test period.
- Provide clear justification for the recommended ARIMA specification.
- Consider potential structural breaks or regime changes in the UK labour market.

Overall assessment

Quant 2 has established a strong statistical foundation for the project. The methodology and diagnostic framework provide a solid basis for applying ARIMA and ARIMAX to the real UK vacancy data in Week 2.

Ready for next stage

Well-prepared to apply the methodology to the real data and recommend a model specification.

Quant 2 — Week 2 Assignment

ARIMA Model Selection, Statistical Diagnostics and Methodological Report

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1 Objectives for Week 2

Select and justify an ARIMA specification using the UK vacancy data.

Apply the methodology to real data

Use the cleaned ONS vacancy series from Quant 1 to estimate and compare candidate ARIMA models.

Conduct statistical diagnostics

Assess stationarity, residual autocorrelation and overall model adequacy.

Recommend a model

Provide a clear justification for the selected ARIMA specification based on statistical evidence and diagnostic tests.

Prepare documentation

Summarise the methodology, results and recommendations in a short report with reproducible code.

2 Specific Tasks

Key tasks to complete for Week 2.

Task	Details
Explore stationarity	• Apply ADF and KPSS tests to the ONS series. • Assess whether differencing is required. • Present plots of the series, ACF and PACF.
Compare ARIMA specifications	• Estimate a small set of candidate models (e.g. different p, d, q values). • Compare using AIC and BIC (or AICc). • Justify the chosen model(s).
Assess diagnostics	• Analyse residuals (ACF, PACF). • Perform Ljung–Box test for autocorrelation. • Check normality of residuals. • Comment on any model limitations.
Prepare final recommendation	• Select a preferred ARIMA specification for use in forecasting (training data only). • Provide clear justification and interpretation of model parameters. • Outline any alternative models considered.

3 Expected Outputs

Files and materials to be submitted by Monday.

Output	Description	Format
Exploratory analysis notebook/script	Code for stationarity tests, model estimation and diagnostics.	Python script (.py or .ipynb)
Model comparison table	AIC/BIC values for candidate models.	CSV or table (in report)
Diagnostic plots	Time series, ACF/PACF, residual diagnostics.	Figures (.png or .pdf)
Methodological report	Summary of approach, results and recommended ARIMA specification.	PDF or Markdown (.pdf or .md)

4 Key Considerations

Points to keep in mind when completing the assignment.

- Use the common ONS dataset with the final 12 observations reserved for out-of-sample testing.
- Model selection must use training data only.
- Keep the model as simple as possible (avoid overfitting).
- Provide clear justification for all choices.
- Highlight any limitations or potential extensions.

Success criteria

A statistically justified ARIMA specification with clear diagnostics and documentation, ready for use in the Week 2 forecasting work by Quant 3.

Deadline

Submit all outputs by **Monday 2 November 2026, 18:00**

Quant 3 — Week 1 Review

Python Implementation, Initial Forecasting Framework and Evaluation

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1 Key Tasks Completed

Developed a working forecasting prototype.

Built ARIMA prototype
Implemented an initial ARIMA(1,1,1) model using synthetic monthly data.

Naive benchmark
Implemented a naive forecasting benchmark for comparison.

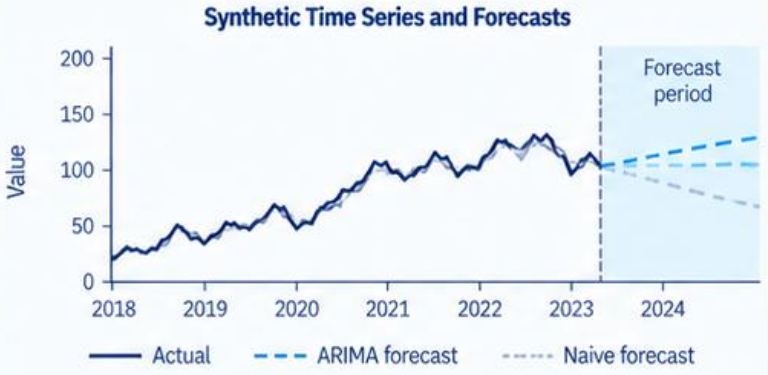
Forecasting metrics
Calculated key performance metrics (MAE, RMSE and MAPE) and compared model performance.

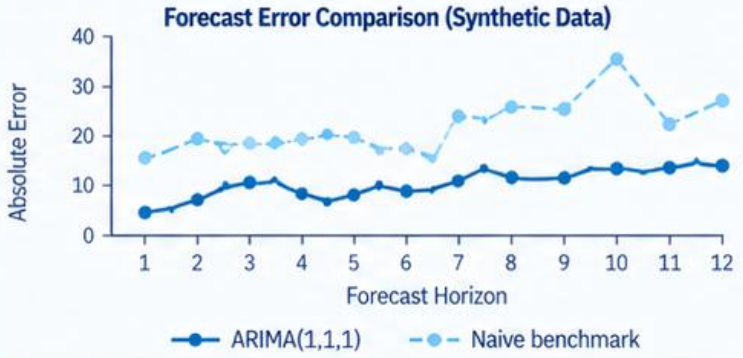
Code structure and documentation
Produced clear, modular Python code with comments and initial documentation.

Initial visualisations
Generated forecast plots and error comparison figures using synthetic data.

2 Illustrative Example (Synthetic Data)

Demonstrating the initial forecasting framework.





3 Strengths and Development Points

Assessment of Week 1 performance.

Key strengths

- Successfully implemented a working forecasting framework.
- Produced clear, modular and well-structured Python code.
- Included an appropriate naive benchmark for comparison.
- Calculated and interpreted key forecasting metrics.
- Produced clear visualisations and outputs.
- Demonstrated good understanding of forecasting evaluation.

Development points

- Apply the framework to the real ONS vacancy series in Week 2.
- Implement a rolling one-step-ahead forecasting evaluation.
- Ensure the code is flexible to incorporate the ARIMA specification from Quant 2.
- Consider additional visualisations (e.g. forecast vs actual, error distribution).
- Document the code for reproducibility.
- Align the evaluation period with the team's agreed train/test split.

Overall assessment
Quant 3 has established a strong computational foundation for the project.
The existing code provides a solid base to apply to real UK vacancy data in Week 2.

Ready for next stage
The forecasting framework is in place and can now be applied to the cleaned real dataset and integrated with the model specification from Quant 2.

Quant 3 — Week 2 Assignment

Apply the forecasting framework to real UK vacancy data and evaluate performance.

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1 Objectives for Week 2

Implement real-data forecasting and evaluation.

Apply the existing framework to real data

Use the cleaned ONS vacancy dataset from Quant 1 and implement the forecasting framework.

Produce out-of-sample forecasts

Generate rolling one-step-ahead forecasts for the final 12 observations using the agreed model specification (from Quant 2).

Compare against a naive benchmark

Implement a naive forecast and compare performance using consistent accuracy metrics.

Visualise and document results

Produce clear plots, tables and a short report, with reproducible code.

2 Specific Tasks

Key tasks to complete for Week 2.

Task	Details
Load and prepare data	• Use the final aligned ONS dataset provided by Quant 1. • Check for any remaining missing values or outliers. • Ensure correct date handling and frequency.
Implement rolling forecasts	• Produce one-step-ahead forecasts for the final 12 observations. • Use the ARIMA specification provided by Quant 2 (keep order configurable). • Follow a consistent evaluation precedure (train on initial period, forecast through test period).
Compare models	• Calculate MAE, RMSE and MAPE for ARIMA and naive benchmark. • Create a table of results. • Analyse the forecast errors.
Produce outputs	• Generate a forecast vs actual plot. • Create an error comparison figure. • Write a short technical report explaining the approach and results.

3 Expected Outputs

Files and materials to be submitted by Monday.

Output	Description	Format
Forecasts file	Actual values, ARIMA forecasts and naive forecasts for the test period.	forecasts.csv
Accuracy metrics	MAE, RMSE and MAPE for ARIMA and naive models.	accuracy.csv
Visualisations	Time series plot and forecast error comparison figure.	Figures (.png or .pdf)
Python script	Updated, fully documented forecasting code.	**forecasting.py** (.py or .ipynb)
Short report	Summary of approach, results and key observations.	Report (.pdf or .md)

4 Key Considerations

Points to keep in mind when completing the assignment.

- Use the same dataset and test period as Quant 2 (final 12 observations).
- Ensure no information from the test period is used in model estimation.
- Keep the code modular and well-documented.
- Make the model specification configurable (do not hard-code the order).
- Clearly compare ARIMA and naive using the same evaluation metrics.
- Provide interpretation of the results, including any limitations.

Success criteria

A functioning real-data forecasting framework with clear evaluation results and reproducible code, ready for integration with Quant 1 and Quant 2 in Week 3.

Deadline

Submit all outputs by **Monday 2 November 2026, 18:00**

Integration, Deliverables & Next Meeting

Three specialised contributions, one research project.

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1 Individual Outputs

Three workstreams producing core components.

Quant 1

Economic data and preparation.

- ONS vacancy dataset
- Indeed job-posting dataset
- Aligned dataset
- Data preparation script
- Data dictionary

Quant 2

Statistical modelling and methodology.

- Stationarity analysis
- ARIMA model comparison
- Diagnostic test results
- Recommended specification
- Methodological report

Quant 3

Python implementation and forecasting.

- Real-data forecasting code
- Rolling one-step-ahead forecasts
- ARIMA vs naive evaluation
- Accuracy metrics and figures
- Short technical report

2 Integration Plan

Combine the three components into a single pipeline.

Clean, aligned dataset
Use the final dataset from Quant 1 as the common input.

↓

Model specification
Apply the ARIMA specification recommended by Quant 2.

↓

Forecasting and evaluation
Implement the forecasting framework from Quant 3 using the agreed setup.

↓

Joint outputs
Integrated results, figures and discussion for the research paper and GitHub repository.

3 Next Steps

From Week 2 deliverables to Week 3 ARIMAX development.

1. **Review Week 2 outputs (end of Week 2)**
   Check data, model specification and forecasting results.
2. **Integrate components (start of Week 3)**
   Combine datasets, ARIMA model and forecasting code into a single reproducible workflow.
3. **Begin ARIMAX development (Week 3)**
   Incorporate Indeed job-posting data and compare ARIMA vs ARIMAX forecasting performance.
4. **Continue with analysis and robustness (Weeks 4–5)**
   Evaluate results, test robustness and interpret findings.
5. **Finalise paper and repository (Week 6)**
   Complete written paper, figures and fully documented GitHub repository.

Next meeting

Review Week 2 submissions and plan Week 3.

Monday 2 November 2026, 18:00 — Online (link to be shared)

Overall goal

A high-quality, reproducible research project that investigates whether Indeed job-posting data can improve forecasts of UK vacancies, resulting in a jointly authored research paper and public GitHub repository.